```
GET networks/_search
{
  "query":{
    "term": {
      "router_ip": { "value": "35.177.0.0/16" }
    }
  }
}
```

Term-level search for IP addresses

Searches for IP addresses in this range

The previous couple of sections have reviewed the core data types. Elasticsearch provides a rich set of data types for almost any use case we can think of. Some core data types are straightforward and intuitive to work with, while others, like `object`, `nested`, `join`, `completion`, and `search_as_you_type` require special attention. In the next section, we explore a few of these advanced data types.

4.6 Advanced data types

We've looked at core and common data types to represent data fields. Some can be classified as advanced types, including some specialized types. In the next few sections, we touch on these additional types with definitions and examples.

> **NOTE** Covering all the data types would bloat this book, and I am also not a big fan of overly long chapters. So, I've made the judicious call to include only the most important and useful advanced data types. The book's GitHub repository and website include examples of the types cited here as well as those omitted.

4.6.1 The geo_point data type

With the advent of smartphones and devices, location services and searching for nearby places have become common. Most of us have used a smart device to do things like find the location of the nearest restaurant or asked for GPS directions to our mother-in-law's house at Christmas. Elasticsearch has a specialized data type to capture a place's location.

Location data is expressed as a `geo_point` data type representing longitude and latitude. We can use this to pinpoint an address for a restaurant, school, golf course, and so on. Let's see it in action.

The following code shows the schema definition of a `restaurants` index: restaurants with `name` and `address` fields. The only notable point is that the `address` field is defined as a `geo_point` data type.

Listing 4.20 Mapping schema with a field declared as a `geo_point` type

```
PUT restaurants
{
  "mappings": {
    "properties": {
      "name":{
        "type": "text"
```